

CS543 WEB SERVICES — ASSIGNMENT 3

Integrating an External SOAP Partner (UniPay Gateway)

Team ID / No: 5

Team Leader: Sushant prashant Prabhavalakar | **ID:** 20251651093

Team Member: Harsh Yadav | **ID:** 20251651042

1. CONTEXT & PARTNER CHOICE JUSTIFICATION

CampusEats integrates with an external bank gateway, **UniPay**, to handle direct card charges for student transactions. While the internal architecture of CampusEats is designed around lightweight REST microservices, SOAP is specifically selected for this external banking edge because UniPay requires a rigid contract-first protocol.

SOAP provides a formal, strongly-typed XML Schema contract via WSDL to guarantee message correctness and prevent malformed transaction calls. Furthermore, financial transactions demand enterprise-grade message-level security (WS-Security) to sign and encrypt request envelopes end-to-end across transport boundaries, alongside standardized transactional reliability (\$WS-^*\$ stack) for payment authorization.

Integrated Operation: charge — Accepts a payment token (cardToken), numerical monetary amount, and currency code, returning a payment status and transaction ID upon execution.

2. HTTP BINDING SPECIFICATION

The SOAP request is carried synchronously over an HTTP POST connection directed to the endpoint defined in the PaymentsService WSDL service/port. The exact HTTP request header and envelope binding block are defined below:

```
POST /pay HTTP/1.1
Host: api.unipay.example
Content-Type: text/xml; charset=utf-8
SOAPAction: "urn:unipay:payments/charge"
Content-Length: 482

<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
    xmlns:uni="urn:unipay:payments">
  <soap:Header>
    <uni:authHeader>
      <uni:merchantKey>CE-MERCHANT-KEY-994821</uni:merchantKey>
    </uni:authHeader>
  </soap:Header>
  <soap:Body>
    <uni:charge>
      <uni:cardToken>tok_9f2c7182a</uni:cardToken>
      <uni:amount>300.00</uni:amount>
      <uni:currency>INR</uni:currency>
    </uni:charge>
  </soap:Body>
</soap:Envelope>
```

3. SERVICE DISCOVERY & REGISTRY ENTRY

In modern enterprise integration practice, service discovery operates through direct contract delivery or structured service catalogues (modern UDDI-style registry entries). CampusEats obtains the UniPay contract either via a direct secure URL (`https://api.unipay.example/Payments.wsdl`) provided during partner onboarding, or by querying an enterprise API registry.

The formal tModel-style registry entry for UniPay in the CampusEats internal catalog is structured as follows:

ENTERPRISE CATALOGUE / REGISTRY ENTRY

Business:

UniPay Bank Ltd.

Service:

PaymentsService

Endpoint:

https://api.unipay.example/pay

tModel Spec:

urn:unipay:payments (WSDL 1.1 Document/Literal)

WSDL Location:

https://api.unipay.example/Payments.wsdl

4. EXTERNAL FAULT MAPPING

To preserve service boundaries and maintain standard error contracts across CampusEats, external SOAP details and bank-specific fault vocabularies must never leak to students or client applications.

When UniPay returns a structured `soap:Fault` (such as a `CARD_DECLINED` code), the internal CampusEats **Payments** service acts as an abstraction layer. It catches the XML fault, extracts the fault code, and maps it directly into the standardized domain error promised in the CampusEats API contract (Assignment 2).

UniPay SOAP Fault Element	Internal Mapping Layer	CampusEats API Contract Response
<code>faultcode: soap:Client</code> <code>errorCode: CARD_DECLINED</code>	→ Payments Service Boundary →	<code>placeOrder: "payment declined"</code> <i>(Status 400 Bad Request / Domain Error)</i>
<code>faultcode: soap:Server</code> <code>errorCode:</code> <code>GATEWAY_UNAVAILABLE</code>	→ Payments Service Boundary →	<code>placeOrder: "payment gateway unavailable"</code> <i>(Status 503 Service Unavailable)</i>

By maintaining this strict boundary mapping, CampusEats guarantees that internal services and mobile clients interact solely with native CampusEats domain errors, fulfilling the contract established in Assignment 2.